FIELD INTELLIGENCE SYSTEM

PHANTOM

Strategic Proposal for Executive Leadership

Standardizing Operational Excellence Across GPU Pod Deployments

~16,300	70+	338	100%
LINES OF CODE	DEPLOY FUNCTIONS	TOTAL FUNCTIONS	OFFLINE CORE

Prepared by John Hamilton

Data Center Technician · CoreWeave AUS-01

darkmatter024.github.io/phantom/dct-ios.html

EXECUTIVE SUMMARY

PHANTOM is a field intelligence system purpose-built for CoreWeave's Data Center Technicians. It exists to solve a singular, measurable problem: the knowledge required to execute GPU pod deployments, triage infrastructure faults, and maintain operational continuity does not persist between shifts, between technicians, or between facilities.

Today, critical operational data — triage sequences, fix histories, deployment states, optic inventories, and escalation thresholds — exists in Confluence pages no one reads, Slack threads no one searches, and the heads of technicians who may not be on shift tomorrow. PHANTOM eliminates this institutional dependency by consolidating field intelligence into a single, deterministic, offline-capable system that operates at the point of execution: the rack.

The system is not a prototype. At ~16,300 lines of production code, 338 functions, and a fully instrumented deployment pipeline with role-based access control, audit trails, and auto-generated completion reports, PHANTOM represents an operational standard — not a concept.

CORE VALUE PROPOSITION

- > Bridges the gap between institutional knowledge silos and real-time floor execution
- > Standardizes triage, deployment, and incident response across every shift and facility
- > Operates 100% offline for core functions — no connectivity dependency on the floor
- > Eliminates redundant rack walks through structured mission briefs and pre-staged intelligence
- > Produces auditable, traceable records of every deployment action from EDP ingest to sign-off
- > Designed compliance-first — deterministic client-side parsing, zero-persistence AI routing, token-gated access

THE COST OF LATENCY

Why knowledge loss is a financial and operational liability.

Every shift change at a CoreWeave facility represents a potential fracture in operational continuity. The technician who diagnosed a failing optic on Rack R-07 at 2:00 AM is not the same technician who arrives at 6:00 AM to complete the replacement. The context — which ferrule was dirty, which port showed the anomalous Rx reading, which spare was already staged — leaves with the outgoing tech.

This is not a training problem. It is an architectural one. The tools available to field technicians were designed for desk-based engineers, not for personnel operating at the rack with gloved hands and limited screen time. The result is a compounding operational tax:

Redundant Rack Walks

A technician reads a ticket, walks to the rack, realizes they lack context or tooling, returns to their workstation. Conservative estimate: 20 minutes per incident. At scale, this represents hundreds of lost labor-hours per quarter.

Knowledge Attrition

When experienced technicians leave — whether for the day or permanently — their accumulated fix patterns, escalation instincts, and facility-specific knowledge leave with them. No system captures it. No system surfaces it to their replacement.

Inconsistent Execution

Without a standardized triage sequence, two technicians troubleshooting identical faults will follow different procedures, document differently, and escalate at different thresholds. The variance is invisible until it causes an outage.

Deployment Opacity

GPU pod deployments spanning dozens of racks across five sequential phases generate no centralized state. Phase progress, optic consumption, blocker status, and acceptance criteria exist only in the heads of the build team — or not at all.

INTEGRATED INTELLIGENCE

The convergence architecture.

PHANTOM is not a collection of utilities. It is an integrated intelligence system where every feature generates structured data that another feature consumes. The result is a compound learning loop: the system becomes measurably more valuable with every shift completed, every fix logged, and every deployment closed.

THE FEEDBACK LOOP

Triage > Ghost Echo
Structured fault data from the 7-step triage protocol feeds the incident memory engine. Every resolved fix becomes a retrievable pattern.

Ghost Echo > Hypothesis Engine
Accumulated fix records produce confidence-scored action plans. New technicians inherit the field experience of every predecessor.

EDP Parser > Build Lead
Client-side EDP parsing seeds rack records, device inventories, fiber connections, and optic counts directly into the deployment pipeline. Zero API calls.

Build Lead > Task Engine
Each deployment phase generates task checklists from templates. Completion tracking per rack, per phase. Phase gates enforce sequential integrity.

Build Lead > Acceptance Criteria
Six acceptance criteria — three auto-evaluated, three manual sign-off — gate deployment completion. Audit trail captures every state transition.

Deployment > Completion Report
Auto-generated HTML and JSON reports aggregate all deployment data: phases, optics, blockers, triage, audit trail. Exportable and printable.

Voice Assistant > All Systems
Natural language interface with access to the full knowledge base — optic specs, CLI commands, Ghost Echo history, site profile context. Hands-free on the floor.

DEPLOYMENT ENGINE

The 70-function pipeline from EDP ingest to sign-off.

The Build Lead deployment engine is the operational core of PHANTOM. It transforms an Engineering Design Package (EDP) — typically a CSV manifest of rack configurations, device inventories, and fiber connections — into a fully tracked, phase-gated deployment pipeline with optic ledger, role-based access control, and auto-generated completion reports.

FIVE-PHASE SEQUENTIAL PIPELINE

MECHANICAL
Physical rack installation, rail mounting, cable management, labeling verification

POWER
PDU connections, circuit validation, power draw verification, UPS integration

NETWORK
Fiber runs, optic seating, port mapping, fabric connectivity testing

COMPUTE
GPU seating, node bring-up, BIOS/firmware verification, thermal baseline

VALIDATION
End-to-end testing, acceptance criteria evaluation, build lead sign-off

KEY CAPABILITIES

Client-Side EDP Parsing

Deterministic CSV parser runs entirely in-browser. Zero API dependency for structured data extraction. SHA-256 parse caching eliminates redundant processing.

Optic Ledger

Per-type tracking of required, dispensed, and installed optics. Guards prevent installing what hasn't been pulled. Live dashboard with tap-to-adjust controls.

Role-Based Access Control

PIN-gated lead mode with SHA-256 hashed credentials stored per-deployment. Floor techs see all data but cannot complete deployments or modify acceptance criteria.

Multi-Deployment Command Center

Aggregate dashboard across all active deployments. Phase heat map with color-coded cells. Live blocker counts, optic consumption, and completion percentages.

Audit Trail

Append-only event log across 11 action types. Full traceability from EDP ingest to deployment completion. Records are immutable after write.

SECURITY & COMPLIANCE

The CISO pillar: compliance-first architecture.

PHANTOM was designed to avoid the failure mode of most field-originated tools: becoming IT shadow infrastructure that operates outside organizational security posture. Every architectural decision prioritizes auditability, data minimization, and deterministic behavior over convenience.

ENCAPSULATED PROXY MODEL

All AI-powered features route through a Cloudflare Worker proxy at the network edge. API credentials are stored exclusively in Cloudflare's encrypted environment variables — never in the application, never on the device, never in the repository. The Worker enforces origin validation, accepting requests only from the authorized GitHub Pages domain. This architecture ensures that even complete source code exposure reveals zero secrets.

- API keys stored in Cloudflare encrypted environment variables
- Origin-locked — rejects requests from unauthorized domains
- Anthropic and Jira tokens secured in a single proxy layer
- CORS hardening with explicit origin allowlist

ZERO-PERSISTENCE AI ROUTING

AI queries are processed in-memory by the Anthropic API and returned. Anthropic does not store or train on API traffic. No query history, no response cache, and no user data persists on any external server. The data lifecycle of an AI interaction is: device to proxy to Anthropic to proxy to device. At rest, nothing exists outside the technician's local storage.

- No persistent server-side storage of queries or responses
- Anthropic API data handling: processed and discarded
- Client-side EDP parsing — zero API calls for structured CSV data
- Auto-retry on 429/529 with exponential backoff (8s/15s)

ACCESS CONTROL & DATA GOVERNANCE

Role-based access control uses PIN-gated sessions with SHA-256 hashed credentials stored per-deployment. Lead sessions exist in sessionStorage only — they clear automatically when the application closes. All state mutations are instrumented with an append-only audit trail that records actor, action, timestamp, and deployment context.

- PIN-gated RBAC — SHA-256 hashed, per-deployment, sessionStorage scoped
- Append-only audit trail across 11 instrumented action types
- Completed deployments are immutable — no mutation path exists
- Backup import validates referential integrity before restore

ELIMINATING SHADOW IT

Deterministic architecture by design.

The most significant security risk from field-originated tools is not malicious intent — it is well-meaning engineers building solutions that bypass organizational controls because those controls were never designed to accommodate their workflow. PHANTOM addresses this directly:

Deterministic Client-Side Processing

The EDP parser — the system's most data-intensive operation — runs entirely in the browser. CSV parsing, rack seeding, optic counting, and device classification all execute client-side with zero network calls. This eliminates the most common shadow IT vector: sending proprietary data to unauthorized external services.

No Backend, No Database, No Account System

PHANTOM has no server infrastructure beyond the Cloudflare proxy. All persistent data lives in the device's localStorage. There is no user database, no authentication server, no session store. This dramatically reduces the attack surface and eliminates an entire class of vulnerabilities: SQL injection, credential stuffing, session hijacking, and server-side data breaches.

Single-File Deployability

The entire application is one HTML file. There is no build system, no dependency tree, no npm packages. Every line of code is auditable in a single read. Security scanners can evaluate the complete codebase without toolchain setup. This is not a limitation — it is a deliberate architectural choice that prioritizes auditability over developer convenience.

Transparent AI Integration

AI features are clearly gated and labeled. The site profile context block injected into AI prompts is visible and editable by the user. No hidden data collection, no background telemetry, no behavioral analytics. The system does exactly what it appears to do — nothing more.

OPERATIONAL ROI

Measurable value-multipliers.

The return on PHANTOM is not measured in software licensing savings. It is measured in reduced operational latency, preserved institutional knowledge, and the elimination of unstructured human dependency from safety-critical workflows.

Eliminated Rack Walks

KPI: MINUTES RECOVERED PER INCIDENT

A technician who arrives at a rack with a structured mission brief — fault description, suspected cause, required optic, tools to bring, first action, escalation threshold — does not need to return to their workstation for context. Conservative estimate: 20 minutes saved per incident. At 10 incidents per shift across a facility, this represents over 3 hours of recovered labor daily.

Institutional Memory Retention

KPI: KNOWLEDGE PERSISTENCE ACROSS SHIFTS

Ghost Echo transforms individual fix experience into organizational intelligence. A technician on their first shift inherits the accumulated field knowledge of every technician who worked that rack before them. This is not documentation — it is contextual, ranked, confidence-scored decision support at the point of action.

Deployment Velocity

KPI: DAYS FROM EDP INGEST TO SIGN-OFF

The Build Lead engine tracks every rack through five sequential phases with task checklists, blocker tracking, and acceptance criteria. Phase gates prevent skipped steps. The command center provides aggregate visibility across all active deployments. For the first time, deployment state is observable — not assumed.

Onboarding Acceleration

KPI: TIME TO OPERATIONAL COMPETENCE

A new technician using PHANTOM has immediate access to: the triage flowchart (standardized sequence), the hypothesis engine (ranked actions from historical data), the voice assistant (hands-free reference), and the complete offline reference library. The ramp from hire to effective floor technician is compressed by the depth of the system they carry in their pocket.

RECOMMENDATIONS

Proposed path to production readiness.

PHANTOM is operationally complete but organizationally pre-production. The following recommendations outline the path from current state to enterprise-standard deployment, ordered by priority and dependency.

1. Approve Field Validation with Real Site Data at AUS-01

PHANTOM currently operates on public vendor documentation. The system's full value — particularly Ghost Echo, the hypothesis engine, and the deployment pipeline — requires real rack configurations, optic inventories, and port maps. This is the single highest-impact approval action available.

2. Integrate into CoreWeave SDLC Security Pipeline

Migrate the repository into the CoreWeave GitHub organization to enable standard security scanning. The single-file architecture makes this straightforward — the entire codebase is auditable in one read. The developer has no existing CoreWeave GitHub access and is prepared to work within any process defined.

3. Expand to a Second Facility in Single-Device Mode

PHANTOM's single-file, zero-infrastructure architecture means expansion to a second facility has zero marginal infrastructure cost. A technician opens a URL on their phone. The site profile editor allows facility-specific customization without code changes.

4. Confirm SSO Identity Availability for Field Technicians

Stage 10 (multi-device sync) requires authenticated write access to the Cloudflare Worker. The current PIN system is a convenience control, not a security boundary. Server-side identity validation is a prerequisite for any shared-state architecture. This depends on SSO availability for DCT roles.

5. Establish Data Retention Policy

Define organizational retention periods for PHANTOM data categories: audit trail records (recommended: 7 years for compliance), EDP content (3 years), shift handoff records (1 year). This policy should be established before production deployment to ensure consistent data lifecycle management.

PHANTOM

FIELD INTELLIGENCE SYSTEM · STRATEGIC BRIEF · INTERNAL USE ONLY

PHANTOM exists because the people doing the hardest work in the data center deserve tools that were designed for them — not adapted from something else. The foundation is built. The architecture is sound. The path to production is clear. What remains is the organizational decision to invest in the standard that the floor has been waiting for.

John Hamilton

Data Center Technician · CoreWeave AUS-01

linkedin.com/in/johnhamiltonleadtech

darkmatter024.github.io/phantom/dct-ios.html